METRIC

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8 July 1992
SUPERSEDING
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19 April 1991

MILITARY SPECIFICATION

CANISTER, CHEMICAL-BIOLOGICAL MASK: M10A1

This specification is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of canister for a gas mask.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATT'N: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 4240

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STANDARDS

FEDERAL

FED. TEST METHOD STD. NO. 101 - Test Procedures for Packaging Materials

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111 -5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Us. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

5-3-547		Canister, Chemical-Biological, M1OA1 Assembly
26-19-51	-	Machine, Rough Handling, Q1 13.
38-5-296	-	Meter, Chemical Container, Volume Q134
38-6-351-1	-	Indicator, Resistance, Canister, Air Purifier and Chemical Container, Q112
136-14- 170-5	-	Air Leakage Tester, Flow Type, Q211
136-41-600-11	-	All Purpose Apparatus, Gas Life Testing, Q95
136-41 -850-3	-	Test Apparatus Agent Gas Sorption Q216
136-42-102-65	-	Penetrometer, Filter Testing, DOP, Q127
136-44- 1-5	-	Resistance Indicator, Inlet and Outlet Valve, C212
136-45-842	-	Tester, Canister, Shaker, Q261

SPECIAL PACKAGING INSTRUCTIONS (SPT)

P5-3-547 - Canister, Chemical-Biological Mask: Ml OA1

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNTC-ARI-IOT, Rock Island, IL 61299-7210.)

DEPARTMENT OF THE ARMY

ARMY REGULATIONS

AR 50-6 - Chemical Surety Program for Safety and Security Provisions Involving Chemical Surety Material

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(Copies are available from the U.S. Army Publications Distribution Center, 2800 Eastern Boulevard, Baltimore, MD 21220-2896.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODXSS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see &2).

ASTM STANDARDS

D 2867 - Moisture in Activated Carbon
D 3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 5-3-547 or in subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth in the drawings.

3.1.2 Components. All components of the canister shall conform to the specifications and drawings listed on Drawing 5-3-547 and subsidiary drawings.

3.2 Manufacture and assembly. The canister shall be manufactured and assembled as specified on Drawing 5-3-547 and subsidiary drawings listed thereon.

3.3 FiIling. The chemical container shall be filled with not less than 330 milliliters of impregnated charcoal when tested as specified in 4.4.6.2. The hopper shall be gravity or hand filled from the charcoal drum. The chemical container shaIl be free-fa!l sifter filled. Updraft air flow may be used to remove fines from charcoal during the free-fall filling process. The bottom shall be securely attached to the container body as shown on Drawing

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5-3-549. The filter material on the finished chemical container shall form a firm covering over the screening material, free from cuts, wrinkles, tears, or other surface irregularities and shall be securely crimped (see 6.3).

3.4 Charcoal moisture content. The average moisture content of the completed can isters shall be not greater than 4.0 percent by weight and for any single canister shall be not greater than 5.0 percent by weight when tested as specified in 4.4.6.3.

3.5 Inlet valve disk.

3.5.1 Valve leakage. Air leakage of the valve shall be not greater than 250 milliliters per minute when tested as specified in 4.4.6.4.

3.5.2 Valve resistance. The resistance of the valve shall be not greater than 5 mi!limeters of Water when tested as specified in 4.4.6.5.

3.6 Canister resistance. The airflow resistance of the complete canister shall be not greater than 75 millimeters of water at flow rate of 85 liters per minute when tested as specified in 4.4.6.6.

3.7 Canister leakage. The assembled canister (Drawing 5-3-547) shall not leak when tested as specified in 4.4.6.7. A leak is defined as the formation and release of a single bubble within 10 seconds.

3.8 Gas life. The calculated gas life, X - ks [where X = mean gas life, k = 1.8 for first article and k = 2.219 for regular production and s = standard deviation (see 6.4)], shall be not less than the values shown in table I when tested as specified in 4.4.6.8 (see 4.1 .2).

Table X. Gas life

Gas	Minimum life. minutes	
	Before rough handling	After rough handling
Sarin (GB)	230.0	—--
Cyanogen chloride (CK)	40.0	- - -
Dimethylmethylphosphonate (DMMP)	200.0	200.0

3.9 Dioctylphthalate (DOP) aerosol penetration. The canister, both before and after rough handling (4.4.6. 11), shall show a DOP penetration of not greater than 0.025 percent at a flow rate of 32 liters per minute when tested as specified in 4.4.6.9.

3.10 Charcoal dust emission. The calculated dust emission, X + 2.83s (where X = mean value and s = standard deviation), of the canisters shall be not greater than 5 milligrams (mg) and the dust emission from any individual canister shall be not greater than 5 mg when

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tested as specified in 4.4.6.10. After rough handling (4.4.6.11), the calculated dust emission shall be not greater than 10 mg and the dust emission from any individual canister shall be not greater than 10 mg.

3.11 Rough handling. The canister body shall not have any breaks or open scams, which will allow grit or other foreign matter into the canister, when tested for 15 minutes as specified in 4.4.6.11.

3.12 First article. Prior to the start of regular production, a first article sample (see 6.2) shall be produced in accordance with this specification for examination and test in accordance with 4.3.

3.13 Workmanship. The canisters shall be free from any cracks, rupture seam, defects, contamination or damage.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Governnment. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and S. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Government responsibility. The Government will be responsible for conducting the GB gas life test specified in 4.4.6.8 (see 6.5).

4.1.3 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

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4.1.4 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such altermative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.6.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the co"ntractor's quality program or detailed inspection system, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Inspections to be performed. As determined by the Government, the first article sample items shall be examined and tested in accordance with this specification using either the first article procedures in this specification or other first article procedures which will be provided by the Government. As determined by the Government, the sample first article items may be subjected to any or ail of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings and special packaging instructions.

4.3.2 Sample. The first article sample shall consist of 204 canisters manufactured and packaged (when military packaging is required) using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.3.3 Inspection procedure.

4.3.3.1 For examination. Each canister in the first article sample shall be examined in accordance with the major characteristics of 4.4.5(a), (b), and (c). Fifty randomly selected canisters shall be examined in accordance with the minor characteristics of 4.4.5(a), (b), (c) and, when applicable, the characteristics of 4.4.5(d). Twenty each of every component and subassembly, randomly selected, shall be examined for the requirements of all drawings applicable to each component and subassembly.

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4.3.3.2 For tests. First article sample canisters shall be tested in accordance with table II

Table II. First article tests

Requirement	Sample size	Test paragraph
Volume of filling (3.3) and charcoal moisture content (3.4)"	15	4.4.6.2 4.4.6.3
Valve leakage (3.5.1)	204	4.4.6.4
Valve resistance (3.5.2)	204	4.4.6.5
Canister resistance (3.6)	204	4.4.6.6
Canister leakage (3.7)	204	4.4.6.7
Gas life (3.8)* CK before rough handling DMMP Before rough handling After rough handling GB before rough handling	18 9 9 18	4.4.6.8 4.4.6.8 4.4.6.8 4.4.6.8
DOP aerosol penetration (3.9) Before rough handling After rough handling*	204 20	4.4.6.9 4.4.6.9
Charcoal dust emission (3.1 O) Before rough handling After rough handling*	50 50	4.4.6.10 4.4.6.10
Unit pack container leakage (5.2)	204	4.4.7.1
Unit pack container tear strip (5.2)*	2 2 *	4.4.7.2

l Destructive test.
* l Packaged canisters may be randomly selected from those to be used for charcoal dust emission.

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4.3.4 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the canisters produced by one manufacturer, at one plant, from the same materials, under the same manufacturing conditions, and during one week production, except that no more than one lot of impregnated charcoal or one lot of filter material may be represented in one lot of canisters.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with the classification of characteristics in 4.4.5 and, when specified, table III. Samples shall be selected at random.

4.4.2.2 For test. Sampling shall be conducted in accordance with table IV. Samples shall be selected at random.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Every item in the lot shall be inspected for critical characteristics. The lot represented shall be rejected when nonconformance to a critical characteristic is found. Sample items and, when applicable, the packaging thereof shall be examined in accordance with the classification of characteristics in 4.4.5. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.4.3.2 For test. Sample items and, when applicable, the packaging thereof shall be tested in accordance with table IV. Failure of any sample item to conform to any requirement in table IV based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

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TABLE III. Sampling

Lot size	Inspection levels and sample sizes										
	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11
1201 to 3200	1250	315	125	116	73	53	42	35	23	18	13
3201 to 10000	1250	315	192	116	86	68	50	38	29	22	15
10001 to 35000	1250	315	294	135	108	77	60	46	35	29	15
35001 to 150000	1250	490	294	170	123	96	74	56	40	29	15
150001 to 500000	1250	715	345	200	156	119	90	64	40	29	15
500001 and over	1250	715	435	244	189	143	102	64	40	29	15

*Indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero nonconforming characteristics and reject the lot represented on one or more nonconforming characteristics for ail inspection levels.

4.4.4 Inspection characteristics. Critical characteristics are characteristics whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product or whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are characteristics whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are characteristics whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

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Table IV. Quality conformance tests

Requirement	Sample size and acceptance criteria	Test paragraph
Volume of filling (3.3) and charcoal moisture content (3.4)*	8; Acc 0, Rej 1	4.4.6.2 4.4.6.3
Valve leakage (3.5.1)	Table III, Level IV	4.4.6.4
Valve resistance (3.5.2)	Table III, Level IV	4.4.6.5
Canister resistance (3.6)	100 percent; Acc 0, Rej 1	4.4.6.6
Canister leakage (3.7)	100 percent; Acc 0, Rej 1	4.4.6.7
Gas life (3.8)*		
CK before rough handling	8; Acc 0, Rej 1	4.4.6.8
DMMP		
Before rough handling	8; Acc 0, Rej 1	4.4.6.8
After rough handling	8; Acc 0, Rej 1	4.4.6.8
DOP aerosol penetration (3.9)		
Before rough handling	100 percent; Acc 0, Rej 1	4.4.6.9
After rough handling*	16; Acc 0, Rej 1	4.4.6.9
Charcoal dust emission (3.10)		
Before rough handling	50; Acc 0, Rej 1	4.4.6.10
After rough handling*	50; Acc 0, Rej 1	4.4.6.10
Unit pack container leakage (5.2)	100 percent; Acc 0, Rej 1	4.4.7.1
Unit pack container tear strip (5.2)*	22; Acc 0, Rej 1	4.4.7.2

*Destructive test

4.4.5 Classification of' characteristics. Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on O and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table III for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	5-3-547
3.4.5(a)	Canister, chemical-biological M10A1 assembly		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Specified components correctly assembled	Table III, Level VII	3.1.2, 3.2	VI
102	Solder, paint and varnish evident, correct, and adequate	Table III, Level VII	3.1.1, 3.2	VI
103	Weight marking correct (canister weight shall be within 6 grams (g) of the stamped weight)	Table III, Level VII	3.2	CE
104	Damage	Table III, Level VII	3.13	VI
Minor				
201	Marking evident and legible	Table III, Level IX	3.2	VI
202	Contamination	Table III, Level IX	3.13	VI

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection

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CIASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(b)	Purifier, air assembly		5-3-549
			NEXT HIGHER ASSY 5-3-547

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Specified components correctly assembled	Table III, Level VII	3.1.2	VI
102	Damage	Table III, Level VII	3.13	VI
Minor				
201	Contamination	Table III, Level IX	3.13	VI

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection

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CIASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(c)	Container, chemical assembly		5-3-548
			NEXT HIGHER ASSY 5-3-549

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Specified components correctly assembled	Table III, Level VII	3.1.2	VI
102	Damage	Table III, Level VII	3.13	VI
103	Seam alignment correct	Table III, Level VII	3.1.2	VI
Minor				
201	Contamination	Table III, Level IX	3.13	VI

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.5(d)	Packaging		SPI P5-3-547
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Unit pack assembly correct	Table III, Level IV	5.2	VI and CE
102	Unit pack assembly supplied with tear strip	Table III, Level IV	5.2	VI
103	Corrosion resistant key furnished and attached to end of each can lid as specified	Table III, Level IV	5.2	VI
104	Packing as specified	Table III, Level IX	5.2	VI and CE
105	Unit packing and packing marking evident, correct and legible	Table III, Level IX	5.2	VI and CE

NOTES:
This paragraph is not applicable to packaging for interplant shipment.
CE - Commercial inspection equipment
VI - Visual inspection

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4.4.6 Tests.

4.4.6.1 Chemical Surety Material (CSM) testing precautions and responsibility. This document requires the use and testing of chemical material, including CSM which is potentially y hazardous to per-some]. All applicable safety rules, regulations, and procedures shall be followed in the handling and processing of these materials. Testing with CSM (i.e., CK) shall be conducted in a facility which has been approved for this type of operation by the U.S. Army. All generalized and specialized handling procedures required by Army Regulation AR 50-6 shall be followed when handling or storing CSM (see 6.7).

4.4.6.2 Volume of filling. The volume of filling shall be determined with the Q134 Chemical Container Volume Meter (Drawing 38-5-296).

4.4.6.3 Charcoal moisture content. The charcoal moisture content shall be determined in accordance with ASTM D 2867 using the the oven-drying test method. Samples for moisture content shall be 10 to 12 g.

4.4.6.4 Valve leakage. Valve leakage shall be determined using the Q211 Flow Type Air Leakage Tester (Drawing 136-14-1 70-5).

4.4.6.3 Valve resistance. The valve resistance shall be determined using the Q212 Inlet and Outlet Valve Resistance Indicator (Drawing 136-44-1-5).

4.4.6.6 Canister resistance. The canister resistance shall be determined using the 0112 Air Purifier and Chemical Container Resistance Indicator (Drawing 38-6–35 1-1) as follows: Draw air through the influent opening and out through the effluent opening at a rate equivalent to 85 liters per minute at 25°C and 760 millimeters of mercury (see 6.8 and 6.9). Do not continue the flow of air through the canister any longer than necessary to take resistance reading.

4.4.6.7 Canister leakage. After the canister is assembled, it shall be tested for canister leakage by immersing the item in water at an air pressure differential of 34.5 x 103 Pa (5 psi). Immersion time shall be no less than 10 seconds, commencing after surface and entrapped air bubbles are removed from the item.

4.4.6.8 Gas Life. The CK gas life shall be determined with the Q95 All Purpose Gas Life Testing Apparatus (Drawing 136-41 -600-1 I)(see 4.4.6.1). The DMMP and GB gas life shall be determined with the Q216 Agent Gas Sorption Test Apparatus (Drawing 136-4 1-850-3) (see 6.10). Gas life conditions shall be as shown in table V.

4.4.6.9 Dioctylphthalate [DOP) aerosol penetration. The DOP aerosol penetration shall be determined using the Q127 DOP Filter Testing Penetrometer (Drawing 136-42-102-65).

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TABLE V. Conditions for gas life testing

Gas	Concentration	Temperature	Relative humidity(C)	Rate of airflow
GB	4.0 ± 0.4 mg/liter	26 ± 6°C	Not more than 15%	32.0 ± 1.0 liters/min(A)
CK	4.0 ± 0.4 mg/liter	26 ± 6°C	80 ± 3%	50.0 ± 1.6 liters/min(B)
DMMP	3.0 ± 0.2 mg/liter	26 ± 6°C	Not more than 15%	50.0 ± 1.6 liters/min(A)

(A) Constant flow
(B) Intermittent flow

The gas life shaII be reported to the nearest 0.1 minute.

4.4.6.10 Charcoal dust emission. The dust emission shall be measured using the Q261 Canister Shaker Tester (Drawing 136-45-842). The Q261 shall be used to accelerate dust emission in the canister. While the canister is being shaken, the dust shall be collected on a high efficiency filter which collects particles 0.3 microns or larger by drawing air through the canister at a constant flow rate of 50 + 5 liters/minute for 20 minutes. The dust emission shall be determined gravimetrically, using a balance sensitive to 0.1 mg. The collection filter shall be either moisture-proof or shall be desiccated to ensure minimal moisture absorption.

4.4.6.11 Rough handling. Canisters shall be rough handled using the Q113 Rough Handling Apparatus (Drawing 26-1 9-51). The rough handling machine shall only be operated with four unit packed canisters for not less than 15 minutes.

4.4.7 Packaging tests.

4.4.7.1 Unit pack container leakage. The unit pack container shall be tested in accordance with method 5009.3 of FED. TEST METHOD STD. NO. 101 using the hot water or vacuum chamber technique.

4.4.7.2 Unit pack container tear strip. The key shall be attached to the tear strip tongue of the unit pack can and rotated in a clockwise direction around the circumference of the can. When test is conducted using an empty can, structural integrity of the tear strip area shall be maintained during opening using a lid or other means.

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5. PACKAGING

5.1 Interplant shipment. When packaging is for interplant shipment (see 6.11), the canisters covered by this specification shall be unit packed, packed and marked in accordance with ASTM D 3951.

5.2 Military packaging. For all shipments, other than interplant shipments (see 6.11), packaging (preservation, unit packing, packing and marking) shall be as specified in SPI P5-3-S17 (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that maybe helpful, but is not mandatory.)

6.1 Intended use. This specification covers the Ml OA1 canister which is intended for use as a component of the M24 Aircraft Mask and the M25A1 Tank Mask.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) First article:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

(c) Level of packing required (see SPI P5-3-547)
(d) Palletization required (see SPI P5-3-547)

6.3 Canister closing. The assembly line should be so arranged that the canisters, after filling. are exposed to the atmosphere for the shortest possible time. Immediately after filling and closing, the canisters should be sealed by suitable closures, placed in air tight containers such as charcoal drums, or other adequate containers, to prevent undue moisture absorption, pending placing the canisters in the appropriate packaging container. These precautions are necessary because charcoal can absorb sufficient moisture during the manufacturing operations to cause the canisters to be rejected.

6.4 Standard deviation. To find the standard deviation (s) of a group of results:

Step 1. Square the sum of the individual observations $(EX)^2$.

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Step 2. Multiply the sum of the squares of the individual observations by the number of observations.

Step 3. Subtract step 1 from step 2 and divide by the number of observations multiplied by the number of observations minus one [n(n-l)].

Step 4. Extract the square root of step 3. The process may be indicated thus:

$$s = [(n\Sigma X^2 - (\Sigma X)^2)/(n(n-1))]^{1/2}$$

6.5 Samples for Government testing. Samples for Government testing should be forwarded to Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-QAO-C, Building ES 100, Aberdeen Proving Ground, MD 21010-5423.

6.6 Submission of alternative inspection provisions. Proposed alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.7 Chemical surety material. For additional information on CSM procedures, contact Commander, U.S. Army Chemical Research, Development and Engineering Center, Chemical Surety Office, Aberdeen Proving Ground, MD 21010-5423, telephone number (410) 671-2051.

6.8 Canister resistance. The resistance of the air purifier without filter should not exceed 20 mm of water, and with filter should not exceed 71 mm of water in order to meet the final resistance requirement. The specified resistances are to be measured at a flow rate of 85 liters per minute at 25°C and 760 millimeters of Mercury as stated in 4.4.6.6.

6.9 Airflow resistance. Atmospheric conditions often change rapidly and necessitate recalibration of the airflow resistance measuring device several times a day to ensure correct readings. However, frequent recalibration can generally be avoided by use of a standard resistance for setting the flow manometer at the 85-liter point. If a standard resistance is used it must provide true laminar flow, be properly used to prevent turbulence, and the indicated resistance value checked frequently on a properly calibrated flowmeter.

6.10 DMMP test reagent. DMMP may be purchased from Albright and Wilson, Americas, P.O. Box 26229, Richmond, VA 23260.

6.11 InterPlant shipment (see 5.1). Packaging and marking for interplant shipment is for supplies and materials that do not directly enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor,

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or between contractors and subcontractors, or from a vendor or contractor to a military arsenal, plant or other activity for evaluation, immediate use, or further processing as specified in the applicable contract.

6.12 Changes from previous issues. Marginal notations are not used in this revision. . . to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.13 Subject term (key word) listing.

Charcoal, activated
Cyanogen chloride
Dioctylphthalate
DMMP
Sarin

Custodian:

Army - EA

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-13864B(EA)	2. DOCUMENT DATE (YYMMDD) 920818

3. DOCUMENT TITLE
CANISTER, CHEMICAL-BIOLOGICAL MASK: M10A1

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If Applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME U.S. Army Chemical Research, Development and Engineering Center	b. TELEPHONE *(Include Area Code)* (1) Commercial (301) 671-3259 (2) AUTOVON DSN 584-3259
c. ADDRESS *(Include Zip Code)* Commander U.S. Army CRDEC ATTN: SMCCR-PET-S Aberdeen Proving Ground, MD 21010-5423	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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